



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

Academic Year 2024 – 2025 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. CSE 'B'

Course Code & Title: CS3491 Artificial Intelligence and Machine Learning

Name of the Faculty member (s): Dr.B.Vijayalakshmi, AP (SG)/CSE

Innovative Practice Description

- **Unit / Topic:** Unit I / Search Algorithms

- **Course Outcome:** CO1

- **Topic Learning Outcome:** 1a,1b

- **Activity Chosen:** Problem Based Learning

- **Justification:**

Problem-Based Learning (PBL) enhances the understanding of search techniques by promoting real-world applications, critical thinking, and hands-on experimentation. It helps students analyze algorithmic trade-offs, collaborate, and apply search methods like A* and DFS in AI challenges. PBL fosters problem-solving skills, making AI learning interactive, practical, and industry-relevant.

- **Time Allotted for the Activity:** 20 Minutes

- **Details of the Implementation:**

- The concept of different types of searching algorithms was discussed in a detailed manner.
- After that few example problems were solved using uninformed and informed search techniques.
- The students were formed as a team of 5 members and each team was given a real time problem statement for choosing the search algorithms in the particular scenario along with some more questions.
- The students had a discussion with their peer team members about the given scenario and finally write their findings in the paper
- It is shown in Fig.1. After getting from all the teams, a discussion was made by the faculty member to the entire class.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO1	3	2	1	1	3	1	2	2	2	1	1	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO9	PO10	PSO3
	3	2	1	1	3

Justification for correlation	The students has the knowledge of search algorithms	The students analyze and apply type of search method to use for a particular problem solving.	The students work as a team to find answer for the given task.	The students discuss with the team members and share their ideas.	The concept of searching techniques are essential to find solution for AI based problems
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• Images / Screenshot of the practice:

AI for Medical Diagnosis (Bayesian Networks & Heuristic Search) *Mr. Anurag Sharma*
PA - Nishu
G.K. Sharmasapriya
R. Krishna Praga
R. Manojkumar

★ Scenario:
A healthcare AI system must diagnose diseases based on symptoms provided by patients. The system must efficiently search through a vast medical database to determine the most likely illness.

❓ Question:

1. Why would a Bayesian Network be suitable for handling uncertainty in medical diagnosis?
2. How can heuristic search improve the efficiency of diagnosis compared to brute-force searching?
3. If new diseases emerge, how should the AI system be updated to maintain accuracy?

Ans:

- 1) Bayesian Network is used for uncertainty condition.
Ex: Insulin (independent), Human (dependent)
- 2) As heuristic search gives more information than brute force approach.
↳ Symptoms
- 3) ~~Upd~~ By using and updating learning agents frequently.

AI in Smart Home Automation (Heuristic Search vs. Reinforcement Learning) *Aamitha PK*
Ilavini P
Jane Gracey-D
Afridha Fatima

★ Scenario:
An AI-powered smart home system automatically adjusts room temperatures, lighting, and security settings based on user preferences.

❓ Question:

1. Should the AI use heuristic search or reinforcement learning to optimize energy efficiency? Why?
2. How can AI predict user preferences over time to make better automation decisions?
3. If multiple users have conflicting preferences, how should the AI decide which setting to choose?

- 1) Reinforcement learning. Because, it has to learn and adapt different situations
- 2) Since, it is a learning agent, we have to train the model with many types of scenarios & data to make the model dynamic
- 3) The decision can be made by maximum preferences and rational decision.

2) Web Crawling for Search Engines (BFS vs. DFS)

C. HARI KRISHNAN, N.S. KOTHIRVEL,
M. MEENAKSHI, S. ANANDHAN,

★ Scenario:

A search engine must crawl billions of interconnected web pages to index relevant content efficiently. Some pages contain more valuable information than others, and the crawler must decide which pages to prioritize.

! Question:

1. Should the web crawler use BFS or DFS to explore and index web pages efficiently? Justify your answer.
2. What are the potential challenges of using DFS in this scenario?
3. How can an AI-based heuristic function improve the crawling strategy?

- 1) BFS, since it used to arrange level-by-level manner
- 2) Because of Backtracking search time is more. Inefficient to find the nearest valuable webpage info. Web is enormous so depth will go deep and it's inefficient.
- 3) Using $H(n)$ means the backtracking doesn't happen and the valuable information can't be lost. Therefore improving the backtracking and priority can improve the search.

Fig. 1: Students Answers

• Reflective Critique:

❖ Feedback of practice from students and other stakeholders:

- Students found PBL engaging, improving their understanding of search algorithms through hands-on problem-solving and collaboration.
- Working in groups helped students learn from each other, get answers to questions, and get better at fixing problems.

❖ Benefit of the practice: (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- Increases student motivation and interest by making learning interactive and application-based.

- Develops analytical skills by requiring students to experiment with and compare different search methods.

❖ ***Challenges faced in implementation:***

- Evaluating individual contributions in group-based learning can be challenging.
- Students may struggle with selecting appropriate search algorithms without initial support.

References:

- <https://www.experientiallearningdepot.com/experiential-learning-blog/four-favorite-problem-based-learning-example-activities-to-prioritize-in-your-classroom>
- https://www.ritrjpm.ac.in/images/computer-science/2022-2023/1_GS_CS8592_problem_based_learning.pdf

Signature of Faculty Member

HOD